LEVEL UP ACADEMY

Class 10 | CBSE / GSEB | Science

ANSWER KEY & MARKING SCHEME

Ch 4 Test: Carbon and its Compounds

Chapter / Topic: Carbon and its Compounds

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SECTION A

Q1. Ethane, with the molecular formula C_2H_6, has: **[1]**

Answer: (b) 7 covalent bonds: one C–C bond and six C–H bonds.

Q2. Butanone is a four-carbon compound with the functional group: **[1]**

Answer: (c) ketone

Q3. The next higher homologue of ethanol (C_2H_5OH) is: **[1]**

Answer: (d) C_3H_7OH (propanol). Consecutive members differ by $–CH_2–$.

Q4. While cooking, if the bottom of the vessel is getting blackened on the outside, it means that: **[1]**

Answer: (b) the fuel is not burning completely (the soot is unburnt carbon).

Q5. Which of the following is an unsaturated hydrocarbon? **[1]**

Answer: (a) C_2H_4 (ethene). It has a carbon–carbon double bond.

Q6. **Assertion (A):** Carbon forms covalent bonds rather than ionic bonds. **Reason (R):** Carbon has four valence electrons, and it is difficult for it to gain… **[1]**

Answer: (a). Gaining or losing four electrons needs a lot of energy, so carbon shares electrons instead.

SECTION B

Q7. What is a homologous series? Write the formulae of the first two members of the alkyne series. **[2]**

Answer: A series of compounds in which the same functional group replaces hydrogen in a carbon chain, and consecutive members differ by a $–CH_2–$ unit (14 u). Alkynes: ethyne C_2H_2 and propyne C_3H_4.

- Definition: 1
- Two formulae: ½ + ½

OR, answer: (1) Catenation: carbon can form bonds with other carbon atoms, giving long chains, branched chains and rings. (2) Tetravalency: with a valency of four, carbon can bond with four other atoms of carbon or other elements such as H, O, N, S and Cl.

- Catenation: 1
- Tetravalency: 1

Q8. What is saponification? Write the balanced equation for the reaction of ethyl ethanoate with sodium hydroxide. **[2]**

Answer: The reaction of an ester with an alkali (sodium hydroxide) that gives back the alcohol and the sodium salt of the carboxylic acid. It is used to make soap. $CH_3COOC_2H_5 + NaOH \rightarrow C_2H_5OH + CH_3COONa$

- Definition: 1
- Balanced equation: 1

SECTION C

Q9. Answer the following about ethanol: **[3]**

(i) $2C_2H_5OH + 2Na \rightarrow 2C_2H_5ONa + H_2$. Hydrogen gas is given off and sodium ethoxide is formed. [1]
(ii) Ethene is formed (dehydration): $C_2H_5OH \rightarrow C_2H_4 + H_2O$ [1]

(iii) Ethanoic acid (CH_3COOH) [1]

Q10. What is esterification? Write the balanced equation for the formation of an ester from ethanoic acid and ethanol, and give one use of esters. **[3]**

Answer: The reaction of a carboxylic acid with an alcohol, in the presence of an acid catalyst, to form a sweet-smelling ester. $CH_3COOH + C_2H_5OH \rightarrow CH_3COOC_2H_5 + H_2O$ (with concentrated H_2SO_4). Uses: in perfumes and as flavouring agents.

- Definition: 1
- Balanced equation: 1
- Use: 1

OR, answer: Soaps are sodium or potassium salts of long-chain carboxylic acids. Detergents are generally ammonium or sulphonate salts of long-chain carboxylic acids. In hard water, soap reacts with calcium and magnesium salts to form an insoluble substance called scum, so a lot of soap is wasted. The charged ends of detergents do not form insoluble precipitates with calcium and magnesium ions, so detergents work in hard water too.

- Difference: 1
- Scum formation: 1
- Detergents work in hard water: 1

SECTION D

Q11. Answer the following: **[5]**

(i) A soap molecule has an ionic (hydrophilic) end that dissolves in water and a long hydrocarbon (hydrophobic) tail that dissolves in oil. In water, the tails stick into the oily dirt while the ionic ends face the water, forming clusters called micelles with the dirt in the centre. The micelles stay suspended in water, so the dirt is easily rinsed away. [3]

- Structure of soap molecule: 1
- Micelle formation: 1
- Dirt removed on rinsing: 1

(ii) The addition of hydrogen to unsaturated hydrocarbons in the presence of a catalyst such as nickel or palladium. Use: to make vegetable ghee from vegetable oils. [2]

- Definition with catalyst: 1
- Use: 1

OR, answer: (a) Compounds with the same molecular formula but different structures. Butane C_4H_{10}: n-butane $CH_3–CH_2–CH_2–CH_3$ and isobutane (2-methylpropane) $CH_3–CH(CH_3)–CH_3$. (b) The bonds are formed by sharing electrons, so no ions (charged particles) are formed to carry current.

- (a) definition 1, two structures 1 + 1
- (b) no ions formed 2

SECTION E

Q12. Carbon compounds burn in oxygen to give carbon dioxide and water, along with heat and light. Saturated hydrocarbons generally burn with a clean blue… **[4]**

(i) $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$ + heat and light [1]

(ii) Unsaturated hydrocarbons. [1]

(iii) There is not enough oxygen, so combustion is incomplete and unburnt carbon (soot) is formed. This wastes fuel, blackens vessels, and incomplete combustion can also produce poisonous carbon monoxide. [2]

- Incomplete combustion due to less oxygen: 1
- One problem: 1